

# The skin and systemic disease

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Clues to possible systemic disease:



Rash ass. with joint pain, fever, weight loss, weakness,  
SOB, altered bowel function



Rash not responding to topical tx



Erythema of the skin (inflammation around blood vessels)



Non blanching palpable purplish fixed lesion , painful and blistering >> vasculitis



Unusual changes in pigmentation or texture of the skin.



Palpable dermal lesion 2ry to granuloma , mets , lymphoma and so on...

Characteristic rash that indicates underlying systemic disease :



Erythema multiforme >> HSV, Hep B/C, mycoplasma pneumonia



Pyoderma gangrosum >> IBD , RA , hematological malignancy



Erythema nodosum >> IBD , streptococcal infxn , TB , sarcoidosis , behcet's



Vasculitis >> Hep B /C , SLE , lymphoma , leukemia

Skin reactions ass with infections

1. Toxic erythema



Widespread symmetrical Maculopapular erythema (morbilliform)

■ starts on the trunk then spreads to the limbs



Usually asymptomatic (itchy)



Blanching



Triggered by viruses, bacteria, drugs



Classic presentation??



Tx: No need, can use;

■emollient and topical steroid if symptomatic

■treat the underlying disease

Toxic Erythema

2. EM



Epidemiology



Presentation : target lesions



Pathophysiology: Type IV HSR



Classification : EM Minor vs Major



Causes: according to classification, most common cause: HSV



Dx : Clinical



TX



Recurrence ?



Coarse



Rare associations

EM



The most common infectious trigger HSV



Systemic symptoms of the underlying infection usually precede the EM rash by 2–14 days.

EM Doll's eye Acrallyftandstar

3 shades Dusky red residue necrosis sometimes feet of colors pinkish Red again HSV

5 14 days

Erythema Nodosum



Most common Panniculitis.



Epidemiology: F >> M



Presentation: acute, painful nodules (Shin), especially by palpation.



Idiopathic (50%) vs Secondary



Secondary causes: Multiple, shown in the table



Most Common Sec : Strep



Skin lesions: very tender nodules (3–20 cm), not sharply marginated



Dx & management: Bacterial Culture, Imaging, Dermatopathology



Course: Spontaneous resolution occurs in 6 weeks



Tx: treat secondary; rest + NSAIDs (steroids ?)

Panniculitis adipose tissue inflammation

3. Erythema nodosum Tender subcutaneous nodules Streptococci O over the skin Yersinia multiple bilateral testicular epididymitis A 20-30% idiopathic need to exclude any causes most commonly ifx

EN: Causes

4. Erythema annulare centrifugum

Vs Tinea Corporis cutaneously amenable Yes Tinea pedis idiopathic gpp/fic/hong

5. Erythema chronicum migrans



Caused by Borrelia burgdorferi (Lyme disease)



Migrating erythema



Cutaneous inflammatory response to Borrelia.

Like spirochete like syphilis Transmitted by Ixodes C sign began hematogenous spread Ws Cns Autoimmune like symptoms

Sarcoidosis



Unknown etiology, atypical mycobacterium may be the trigger



Presentation:



May occur with or without cutaneous disease



Skin lesions : Specific vs Non-specific



Earliest: skin-colored papules, on the face, heal without scarring.



Then: Plaques; annular, extremities and Buttocks, SCARRING, Chronic disease



Lupus Pernio: violaceous plaques on the nose and cheeks



EN: m/c NS skin lesion, early sarcoid.



Scarring sarcoidosis.

Sarcoidosis CXR Bilateral hilar lymphadenopathy Noncaseating Curly TB Multisystemic

20 cutaneous Bigmimic

Sarcoidosis Perianths scalyptagy

Dusky infiltrated lesions: Lupus Pernio Badprognosticsign URI IRI osteolytic Acral Those bluish dusky violaceous color

Lupus pernio vs Rosacea midfacial erythema Sharp dermal patient violaceous hue

Scarring sarcoid DDX Tumor hypertrophic pre existing scar sarcoid

Löfgren syndrome

-Erythema nodosum

-Fever

-Arthritis

-Hilar adenopathy

Acute syndrome multisystemic Benign resolves spontaneously

É

Diagnosis and management

■

Imaging

■

Labs

■

Lesional biopsy

■

Tx:

■

Systemic involvement: systemic steroids

■

Limited cutaneous: topical steroids, if extensive: systemic steroids, mtx, hydroxychloroquine.

Skin changes associated with hormonal imbalance

Hyperpigmentation

■

Increase in circulating hormones with melanocyte stimulating activity



Hyperthyroidism, acromegaly, Addison's disease



Pregnancy, OCPs -> melasma/chloasma (localized on the forehead and cheeks)



It may fade slowly if ultraviolet light is excluded from the affected skin using daily sun block.

Generalized Addison melanosis localized creases melasma lentiginosum protection E

Hypopigmentation



Widespread partial loss of melanocyte function



Hypopituitarism (absence of MSH)

Acanthosis nigricans no Black Insulin resistance axilla inguinal neck G Increased thickness of epidermis obese areas extensive involvement Gastric adenocarcinoma

Necrobiosis lipoidica



Associated with DM



Not related to the severity of DM



Coarse not affected by controlling blood sugar



Tx: intralesional steroids, surgery.

destruction of collagen in dermis Atrophy translucent Bx yellowish hue depositing

654 Epp with Check Hgbac

Thyroid disease

edema pitting Orbits shift I

Extensive pitting Grave's disease Generalized myxedema

Graves disease



Hyperthyroidism with diffuse goiter, Ophthalmopathy and dermopathy.



Dermopathy: pretibial myxedema



Tx: steroids

Skin changes with GI and liver diseases



Acrodermatitis enteropathica : genetic disorder of Zinc absorption (seen in neonates)



Acquired zinc deficiency (AZD) occurs in older individuals due to dietary deficiency or failure of intestinal absorption of zinc (malabsorption, alcoholism, prolonged parenteral nutrition)



Skin changes usually appear within weeks of birth with erythematous inflamed scaly skin around the mouth, anus and eyes Zinc deficiency

Breastfed full ATTY of zinc in the transport of zinc like eczema orifices q atrally genetic to therapy mined malabsorption not responding

Zinc def



If the condition is not recognised and treated promptly then the skin can become crusted, eroded and secondarily infected. (candida, S.aureus)



Other findings:



Zinc supplementation (1mg/kg/day), should be continued until zinc levels normalise (or lifelong in inherited forms).

Vitamin C deficiency

■ occurs in those with malabsorption problems, those on a poor diet, the elderly and alcoholics



Vascular fragility



Skin lesions: Petechiae, follicular hyperkeratosis with perifollicular hemorrhage, especially on the lower legs



Other findings:



Labs: Normocytic, normochromic anemia, Serum ascorbic acid level zero. X-ray findings are diagnostic



Tx: Ascorbic acid imp hcollagensynthesis Satb

A Hyperplastic easybleeding afterteethbrushing abnormalhair corkscrew bleeding in hair follicle

3. Pyoderma gangrenosum



Rapid and painful ulcerated necrotic skin areas with hypertrophic undermined purplish margins.



Ass with UC, Crohn's, RA, leukemia, monoclonal gammopathy.

lookfor anycause Tibialtrauma TT papule spushde6

4rapidand\_dden.O PG

4. Dermatitis herpetiformis

Buttonsdisease Celiacdisease extensor surfaces highlypruritic

5. Liver disease

Porphyria cutanea tarda according of whichenzyme ofhememetabolismis a affected theme synthesisdefect and all porphyrias have cutaneous manifested porphyrinsaccumulate Asensitivity to sunlight t recurrentinflame cause scarring milia

Xanthomas Mefull of lipids duetogenetic disease a yellowish lipid O

Hypopigmentation disorders



Albinism , AR , loss of pigment of skin , eyes and hair , no pigment production



Piebaldism , AD , triangular hypopigmented patches , disorder of melanocyte development



Vitiligo localized depigmentation, sharply demarcated, symmetrical macular lesions, loss of melanocytes and melanin



Post- Inflammatory such as psoriasis, eczema, lichen planus and lupus erythematosus; infections, chemicals, reactions to pigmented naevi, genetic diseases

DI D

Congenital there are melanocytes defect'veTsinaseenryme eyeproblemsnystagmus foreheadforetooll Desmith'enofmelanocytes Acquired



Hyperpigmentation disorders There is wide variation in the pattern of normal pigmentation as a result of hereditary factors and exposure to the sun.

Darkening of the skin may be due:



1. An increase in the normal pigment melanin



2. Deposition of bile salts from liver disease



3. Iron salts (haemochromatosis)



4. Drugs or metallic salts from ingestion



5. AN is characterised by darkening and thickening of the skin of the axillae, neck, nipples and umbilicus



6. Post-inflammatory pigmentation is common, often after acute eczema, fixed drug eruptions and lichen planus.



7. In malabsorption syndromes such as pellagra and scurvy, there is commonly increased skin pigmentation.

Skin changes of underlying malignancy

Mycosis fungoides

In cutaneous T cell lymphoma different clinical presentation

Poikiloderma (telangiectasia, reticulate pigmentation)

Hyperpigmentation after radiotherapy

Skin and Pregnancy



Prurigo gestationis- is a benign non-specific pruritic (itchy) papular rash that arises during pregnancy and is generally more severe in the first trimester.



Polymorphous eruption of pregnancy (PEP) is a pruritic erythematous rash that usually starts in the striae of the abdomen during the third trimester and can become widespread. No effect on baby



Pemphigoid gestationis (PG) is a rare disorder that may initially resemble PEP but develops pemphigoid-like vesicles, spreading over the abdomen and thighs (Figure 10.23). PG is an autoimmune disorder, ( Small baby and higher mortality ).

Eigg low weight low gestahage

PEP poly not vesicles eczematous papules duetotension Fina umbilical region no recurrence

Pemphigoid gestations

0 Buttons pemphigoid o periumbilical Baby can be affected t recurrence

GOOD LUCK